

Real-World Problems and Significant Digits

MA.8.NSO.1.6

I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Yesterday you multiplied and divided in scientific notation.

Today the numbers are measurements — and a measurement only knows so much.

Warm-Up

On your own. Four minutes. No notes.

1. $(4 \times 10^5)(3 \times 10^{-2})$
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$
3. Write 0.00902 in scientific notation.
4. Round 4.738 to the nearest hundredth.

Warm-Up

Answers.

1. $(4 \times 10^5)(3 \times 10^{-2})$ **1.2×10^4**
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$ **5.89×10^7**
3. Write 0.00902 in scientific notation. **9.02×10^{-3}**
4. Round 4.738 to the nearest hundredth. **4.74**

Notes

Copy the table.

I. Which digits are significant

Measurement	Digits	Why
4,207	4	the middle zero counts
0.0058	2	leading zeros never count
3.200	4	trailing zeros count after a decimal point
52,000	2	no decimal point, so they do not

Notes

II. Multiplying or dividing: count the digits

$$(2.5 \times 10^2)(4.13 \times 10^2) = 1.0325 \times 10^5$$

2 significant digits and 3, so the answer keeps 2: 1.0×10^5

Multiply or divide: the answer takes the smaller of the two significant-digit counts.

Notes

Copy all three lines.

III. Adding or subtracting: a different rule

$$1,234 + 5.6 = 1,239.6$$

1,234 stops at the ones place; 5.6 stops at the tenths

so the sum stops at the ones place: $1,240 = 1.240 \times 10^3$

Add or subtract: the answer stops at the leftmost of the two last-digit places.

Example 1

A product of two measurements.

A solar array measures 1.25×10^2 m by 8.6×10^1 m.

What is its area?

Give the answer to the correct number of significant digits.

Example 1

Worked.

$$(1.25 \times 10^2)(8.6 \times 10^1) = 10,750 \quad \text{multiply in full}$$

$$1.25 \rightarrow 3 \text{ digits}, \quad 8.6 \rightarrow 2 \text{ digits} \quad \text{the answer keeps 2}$$

$$10,750 \rightarrow 11,000 = 1.1 \times 10^4 \quad \text{round once, at the end}$$

Answer: 1.1×10^4 square metres

Your Turn

Same steps, your numbers. Boards up when done.

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

Your Turn

Answer.

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

108,100, kept to two significant digits

Answer: 1.1×10^5 square metres

Example 2

A difference of two measurements.

A container holds 3.09×10^{-3} cubic feet of liquid.

3.472×10^{-5} cubic feet evaporates.

How much is left? Give the answer with the correct precision.

Example 2

Worked.

$$0.00309 - 0.00003472 = 0.00305528 \quad \text{subtract in full}$$

$$0.00309 \rightarrow \text{stops at } 10^{-5} \quad \text{the leftmost last-digit place}$$

$$0.00305528 \rightarrow 0.00306 = 3.06 \times 10^{-3} \quad \text{round there}$$

Answer: 3.06×10^{-3} cubic feet

Your Turn

Same steps, your numbers. Boards up when done.

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

Your Turn

Answer.

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

3,576, stopped at the tens place

Answer: 3.58×10^3 m

Whiteboards • Question 1 of 9

Boards up on three.

How many significant digits are in 0.00470 ?

Answer it.

Leading zeros, then trailing zeros.

Whiteboards • Question 1 of 9

Answer.

How many significant digits are in 0.00470 ?

4, 7 and the last 0

Answer: 3

Whiteboards • Question 2 of 9

Boards up on three.

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 metres deep.

What is the surface area, to the right number of significant digits?

Answer it.

Two digits each.

Whiteboards • Question 2 of 9

Answer.

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 metres deep.

What is the surface area, to the right number of significant digits?

11,900, kept to two digits

Answer: 1.2×10^4 square metres

Whiteboards • Question 3 of 9

Boards up on three.

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 .

How should the answer be written?

Answer it.

Which count wins?

Whiteboards • Question 3 of 9

Answer.

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 .

How should the answer be written?

the smaller count

Answer: 7.41×10^5

Whiteboards • Question 4 of 9

Boards up on three.

$(2.5 \times 10^3) + (4.71 \times 10^2)$ metres, both measured.

Write the sum with the correct precision.

Answer it.

Where does each one's last digit sit?

Whiteboards • Question 4 of 9

Answer.

$(2.5 \times 10^3) + (4.71 \times 10^2)$ metres, both measured.

Write the sum with the correct precision.

2,500 stops at the hundreds

Answer: 3.0×10^3 m

Whiteboards • Question 5 of 9

Boards up on three.

How many significant digits are in 0.03080 ?

A. 4

B. 3

C. 5

D. 2

Answer it.

Whiteboards • Question 5 of 9

Answer.

How many significant digits are in 0.03080 ?

A. 4

B. 3

C. 5

D. 2

Answer: A — 4

Whiteboards • Question 6 of 9

Boards up on three.

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. $2.6 \times 10^5 \text{ cm}^2$

B. $2.61 \times 10^5 \text{ cm}^2$

C. $26.1 \times 10^4 \text{ cm}^2$

D. $2.6 \times 10^4 \text{ cm}^2$

Answer it.

Whiteboards • Question 6 of 9

Answer.

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. $2.6 \times 10^5 \text{ cm}^2$

B. $2.61 \times 10^5 \text{ cm}^2$

C. $26.1 \times 10^4 \text{ cm}^2$

D. $2.6 \times 10^4 \text{ cm}^2$

Answer: A — $2.6 \times 10^5 \text{ cm}^2$

Whiteboards • Question 7 of 9

Boards up on three.

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

Answer it.

Count both measurements.

Whiteboards • Question 7 of 9

Answer.

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

140 exactly, kept to two digits

Answer: 1.4×10^2 km/h

Whiteboards • Question 8 of 9

Boards up on three.

A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured $9.0 \times 10^1 \text{ m}$.

What did the width measure, to two significant digits?

Answer it.

Divide, then count.

Whiteboards • Question 8 of 9

Answer.

A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured $9.0 \times 10^1 \text{ m}$.

What did the width measure, to two significant digits?

$$3,600 \div 90$$

Answer: $4.0 \times 10^1 \text{ m}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Ellis says: 1,234 has 4 significant digits and 5.6 has 2,
so the sum $1,234 + 5.6$ must be written with 2: 1.2×10^3 .

Explain why he is wrong, and give the correct sum.

Write your answer in sentences.

Which rule is for which operation?

Whiteboards • Question 9 of 9

Answer.

Ellis says: 1,234 has 4 significant digits and 5.6 has 2,
so the sum $1,234 + 5.6$ must be written with 2: 1.2×10^3 .

Explain why he is wrong, and give the correct sum.

products count digits; sums count places

Answer: $1,240 = 1.240 \times 10^3$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S